Visual Perception & Pre-Attentive Attributes

Workshop 1 — Module 3

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Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

1. Pre-attentive visual attributes
2. Feature search vs. conjunction search
3. The Gestalt laws of perceptual grouping
4. Visual encoding channels and their effectiveness
5. Weber's law and just-noticeable differences

Learning Outcome
You will be able to select visual encoding channels that exploit the human visual system rather than fighting against it.

Pre-Attentive Processing

What Is Pre-Attentive Processing?

The visual cortex detects certain properties **before conscious attention** engages:

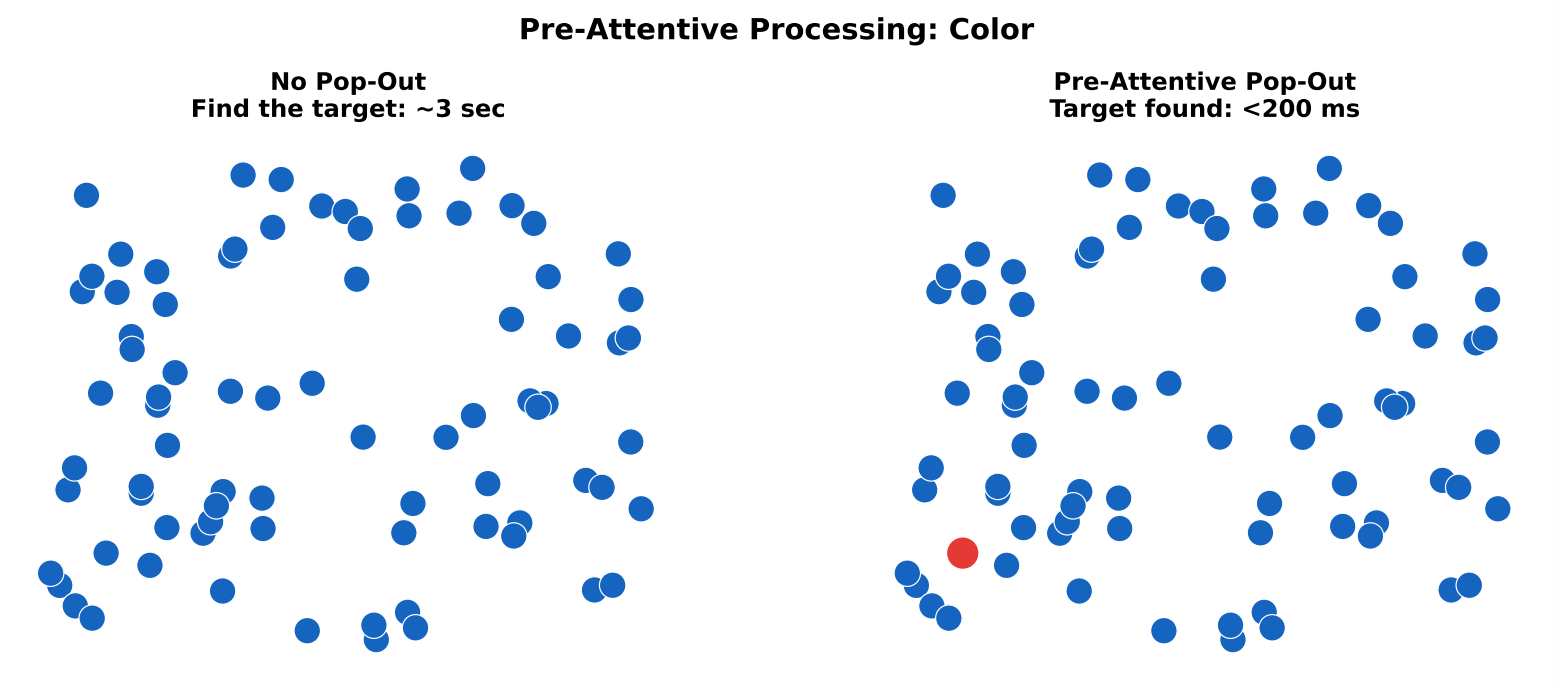
- Processing time: $<$ 200–250 ms
- Independent of the number of distractors
- Driven by the *primary visual cortex* (V1)
- Requires a single distinguishing feature

Four canonical channels:

1. Color (hue)
2. Shape (form)
3. Size (length / area)
4. Orientation (angle)

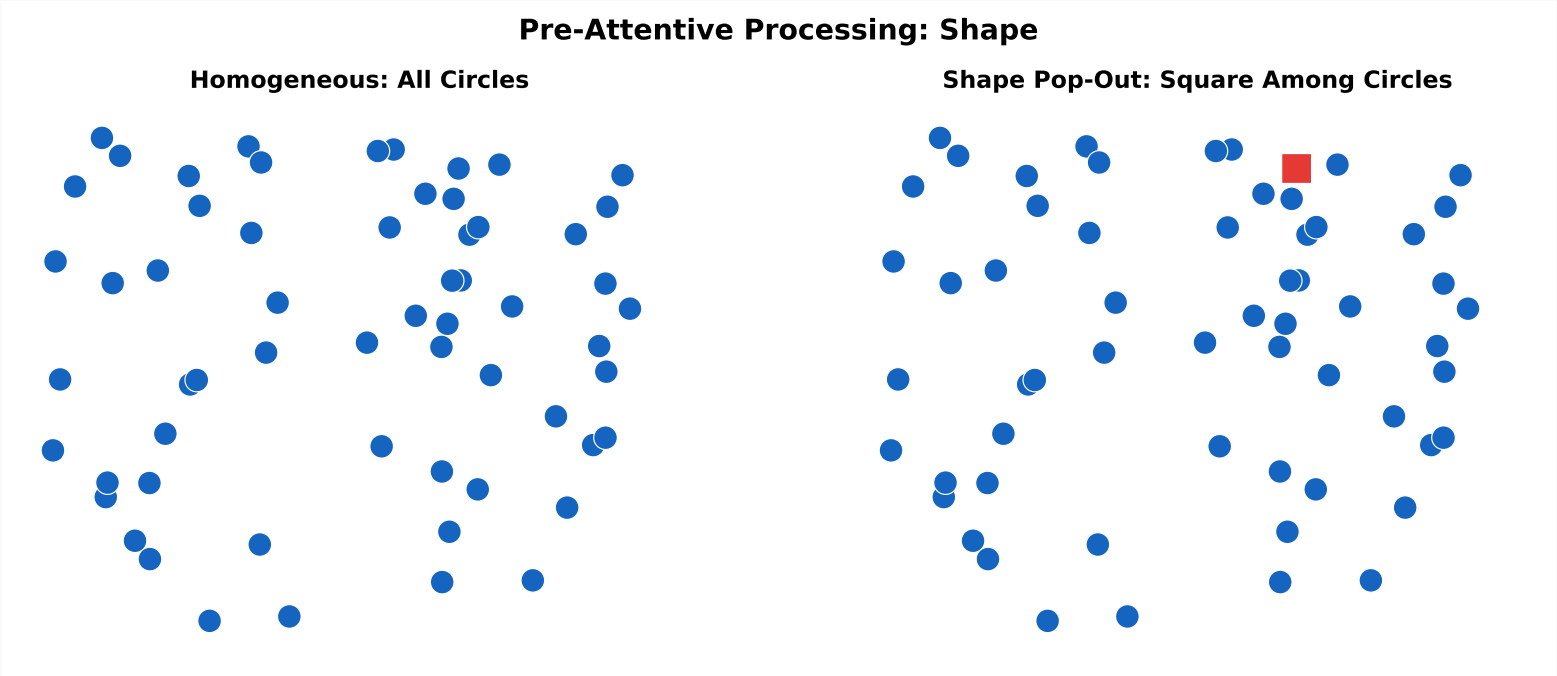
Source: Healey, C. G. & Enns, J. T. (2012). Attention and visual memory in visualization and computer graphics. IEEE TVCG, 18(7).

Pre-Attentive Pop-Out: Color



Pro-Tip
Use colour sparingly: a single accent hue against a neutral field produces instant pop-out. Using > 5 hues destroys the effect.

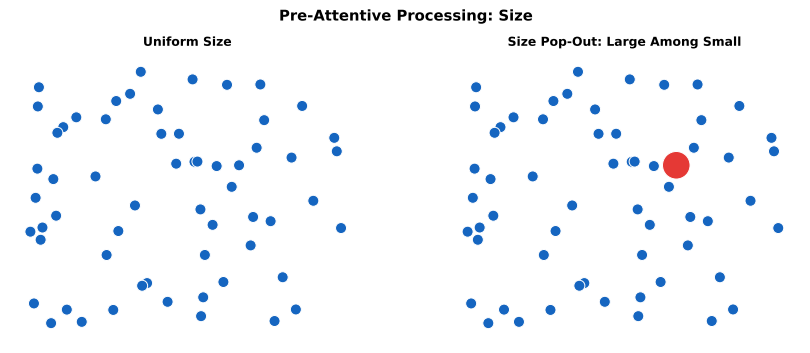
Pre-Attentive Pop-Out: Shape

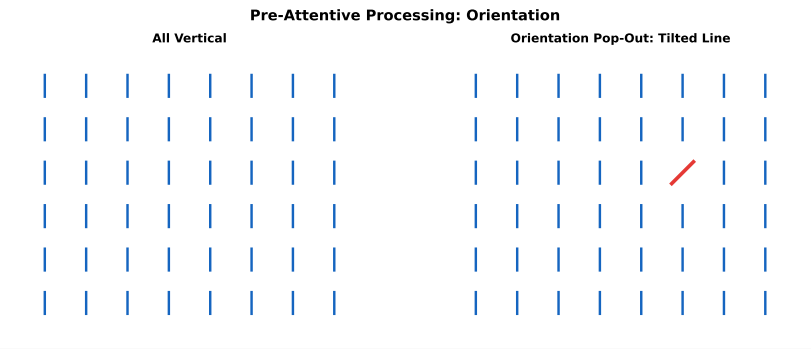


Data Insight

Shape pop-out works only for simple, high-contrast distinctions (circle vs. square). Subtle shape differences (pentagon vs. hexagon) require serial scanning.

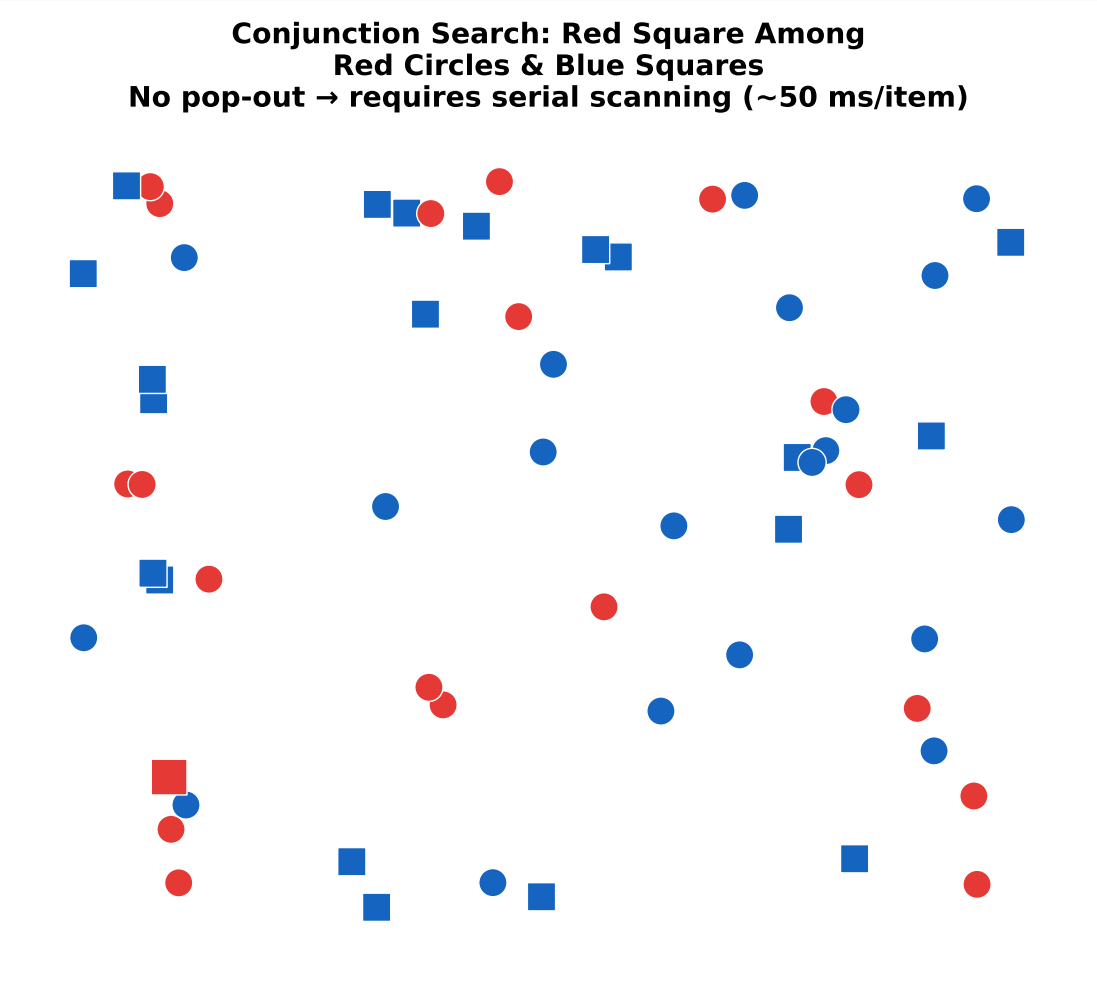
Pre-Attentive Pop-Out: Size & Orientation





Both size and orientation support pre-attentive detection when the difference is sufficiently large. **Weber's law** (later in this module) quantifies "sufficiently large."

Conjunction Search: When Pop-Out Fails



Pro-Tip

When the target is defined by a *combination* of features (red **AND** square), the visual system must search serially at ∼50 ms per item. Avoid encoding critical distinctions as conjunctions — use a single unique

Implications for Chart Design

Do:

- Encode the most important distinction using a single pre-attentive channel (colour or position)
- Limit categorical colours to 5–7 hues
- Use size only for ordinal/quantitative data

Don't:

- Require viewers to distinguish targets by conjunctions (e.g., "find the red dashed line")
- Use shape alone for > 4 categories
- Vary orientation for nominal data

Gestalt Laws of Perceptual Grouping

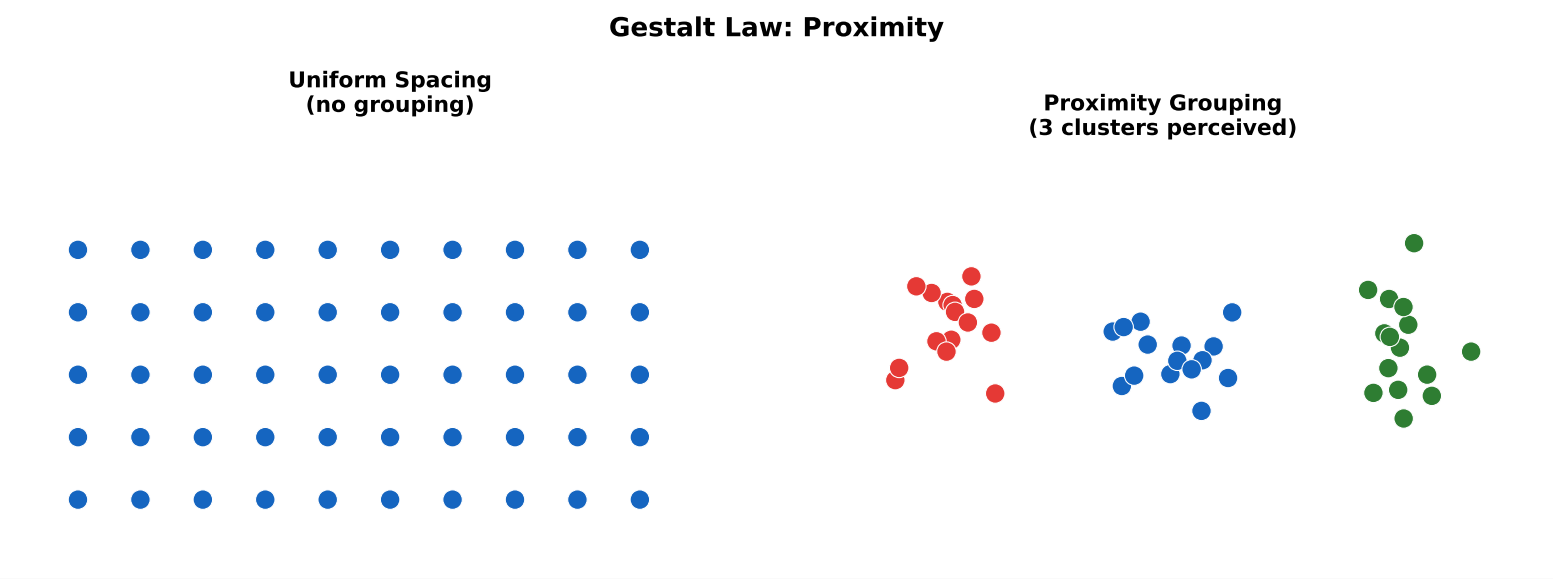
Why Gestalt Matters for Visualization

The Gestalt psychologists (Wertheimer, 1923) identified principles by which the visual system organises stimuli into groups:

1. **Proximity** — near elements form groups
2. **Similarity** — like elements form groups
3. **Continuity** — smooth paths preferred
4. **Closure** — brain completes incomplete forms
5. **Enclosure** — bounded regions create groups
6. **Connection** — linked elements seen as related

Every chart layout implicitly invokes these laws. Effective design *harnesses* them; poor design *violates* them.

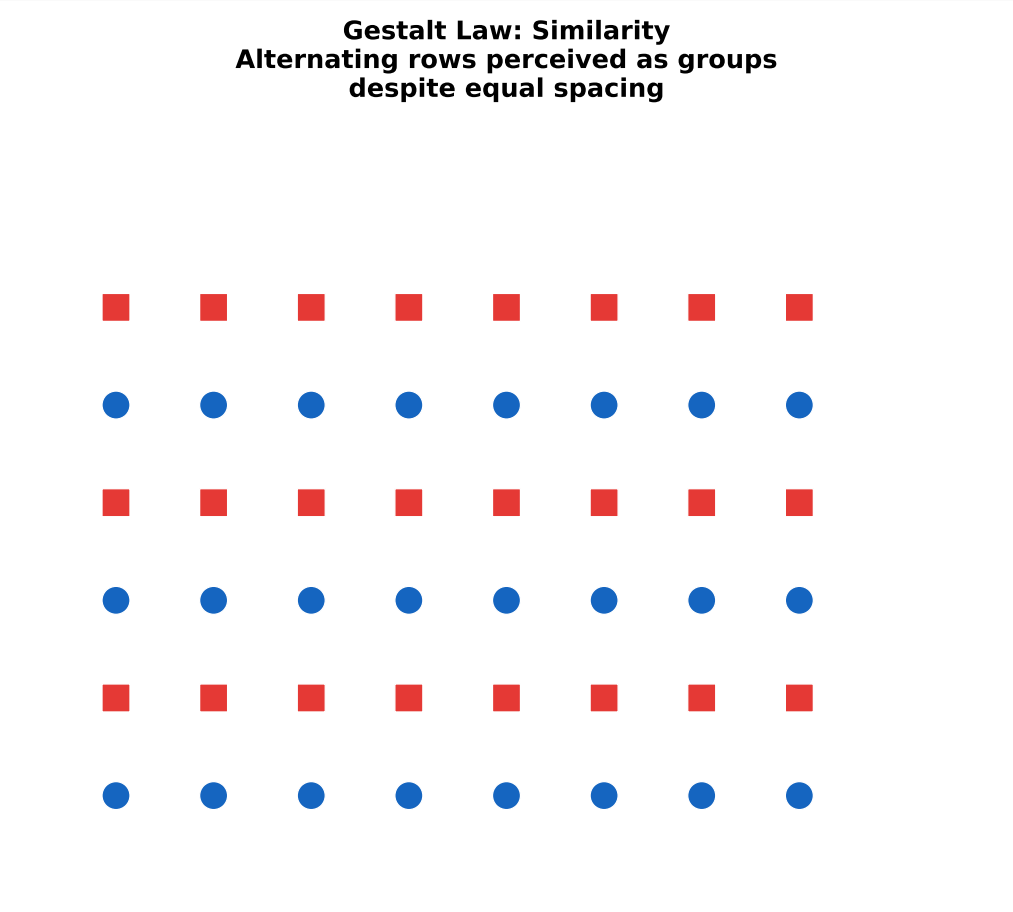
Proximity



Data Insight

Proximity is the strongest grouping cue. In bar charts, spacing between groups must exceed spacing within groups, or the viewer will misread the category structure.

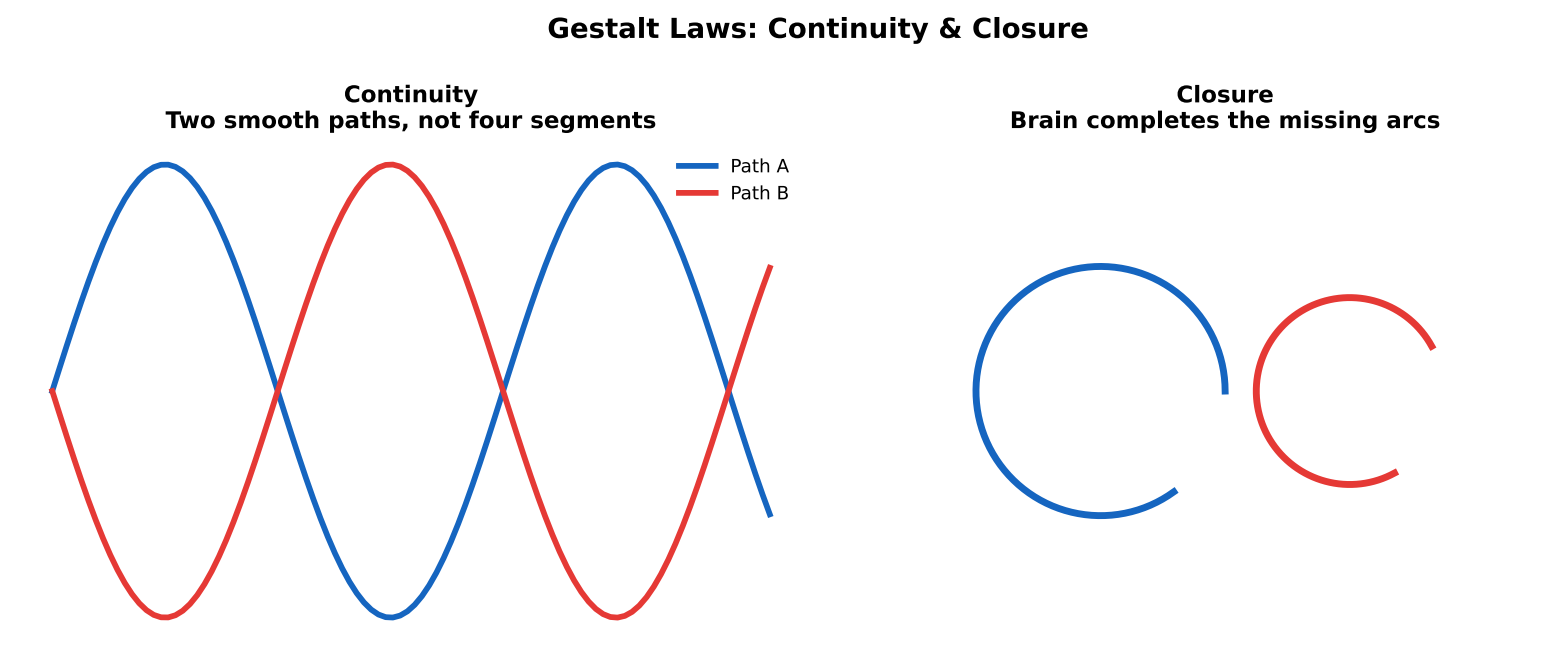
Similarity



Data Insight

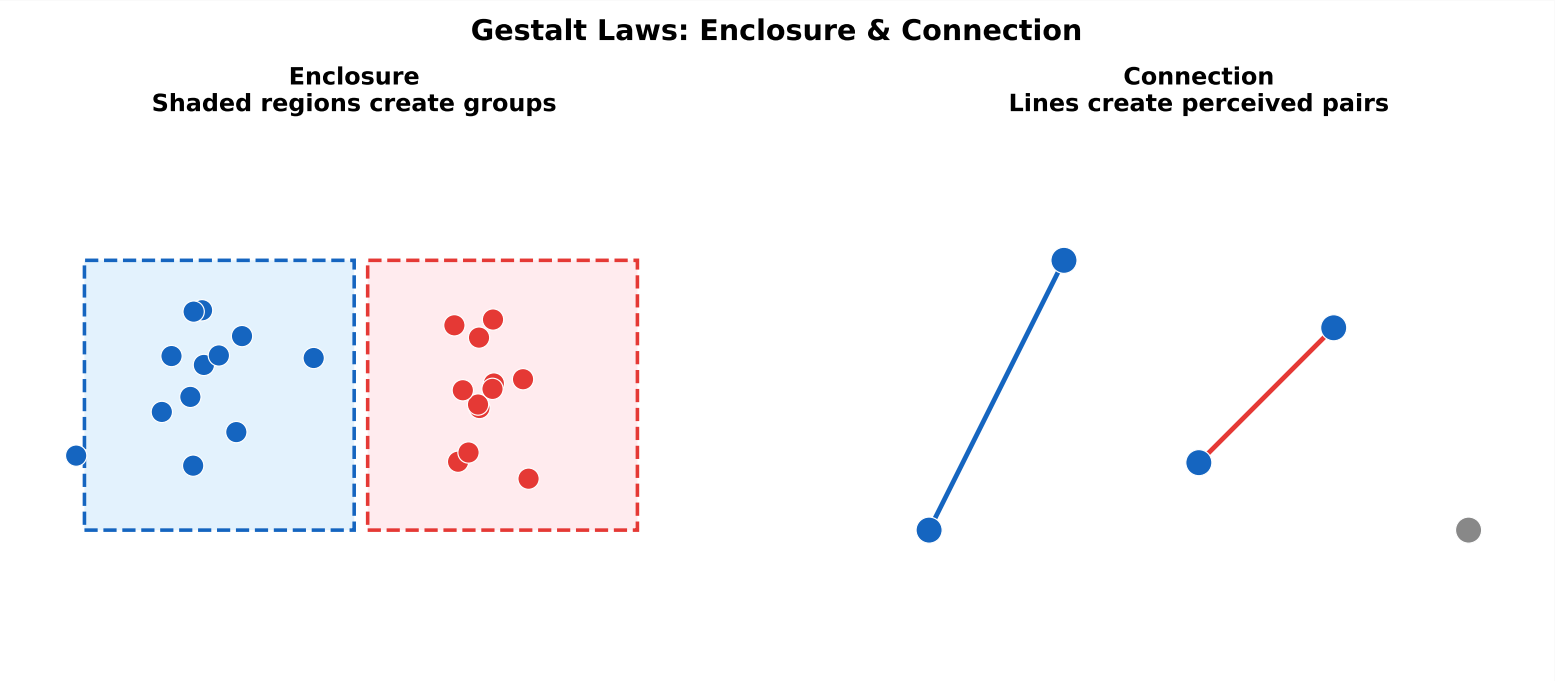
Despite equal spacing, the alternating colour/shape rows are perceived as distinct horizontal groups. This is why colour legends work: similar hues are read as "the same category."

Continuity & Closure



- **Continuity**: two intersecting sine waves are perceived as two smooth paths, not four disjointed segments
- **Closure**: the brain fills in missing arcs to perceive complete circles

Enclosure & Connection



Pro-Tip

Use enclosure (shaded bands) to highlight groups in scatter plots. Use connection (lines between points) to indicate temporal sequence or paired comparisons. Both override proximity when present.

Gestalt Summary Card

The Six Gestalt Laws for Data Visualization

Proximity

Near elements grouped together

Similarity

Like elements grouped together

Continuity

Smooth paths preferred over abrupt changes

Closure

Brain completes incomplete shapes

Enclosure

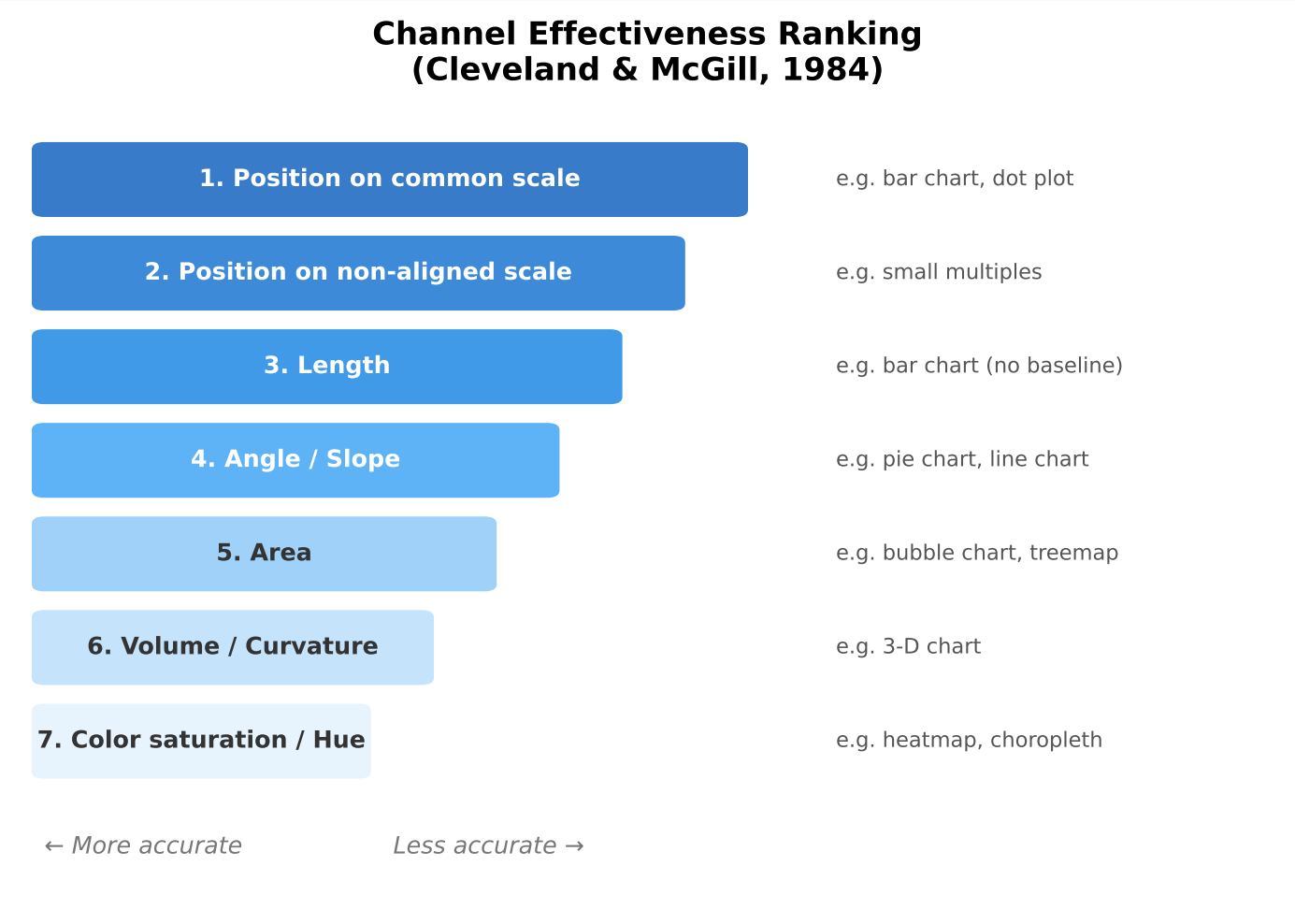
Bounded regions create groups

Connection

Linked elements seen as related

Visual Encoding Channels

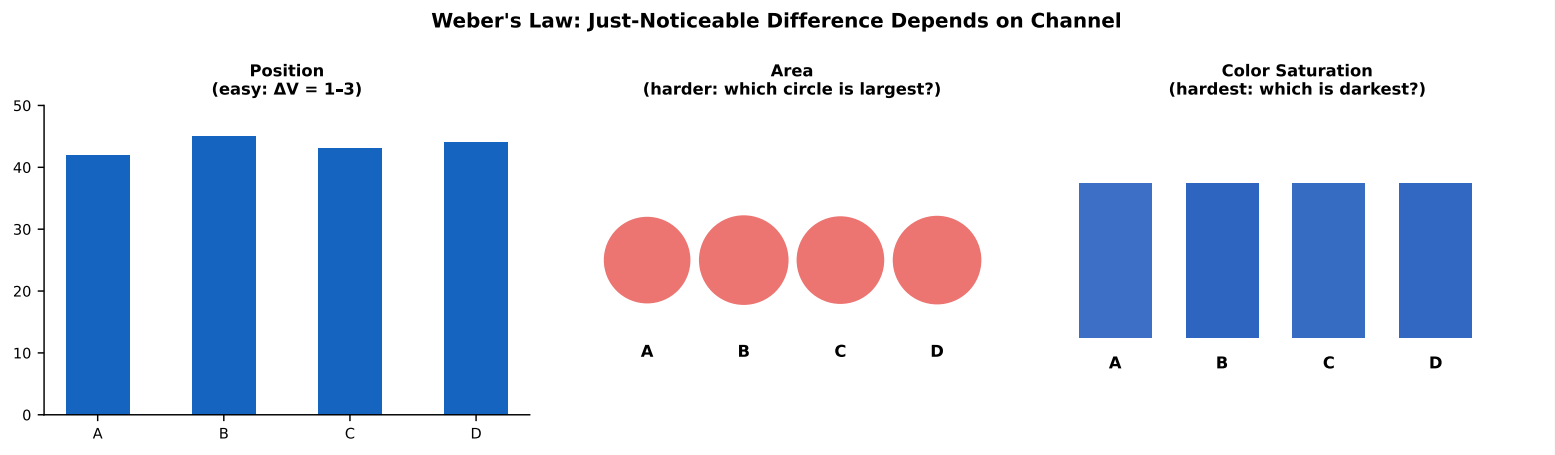
Cleveland & McGill's Channel Ranking



Data Insight

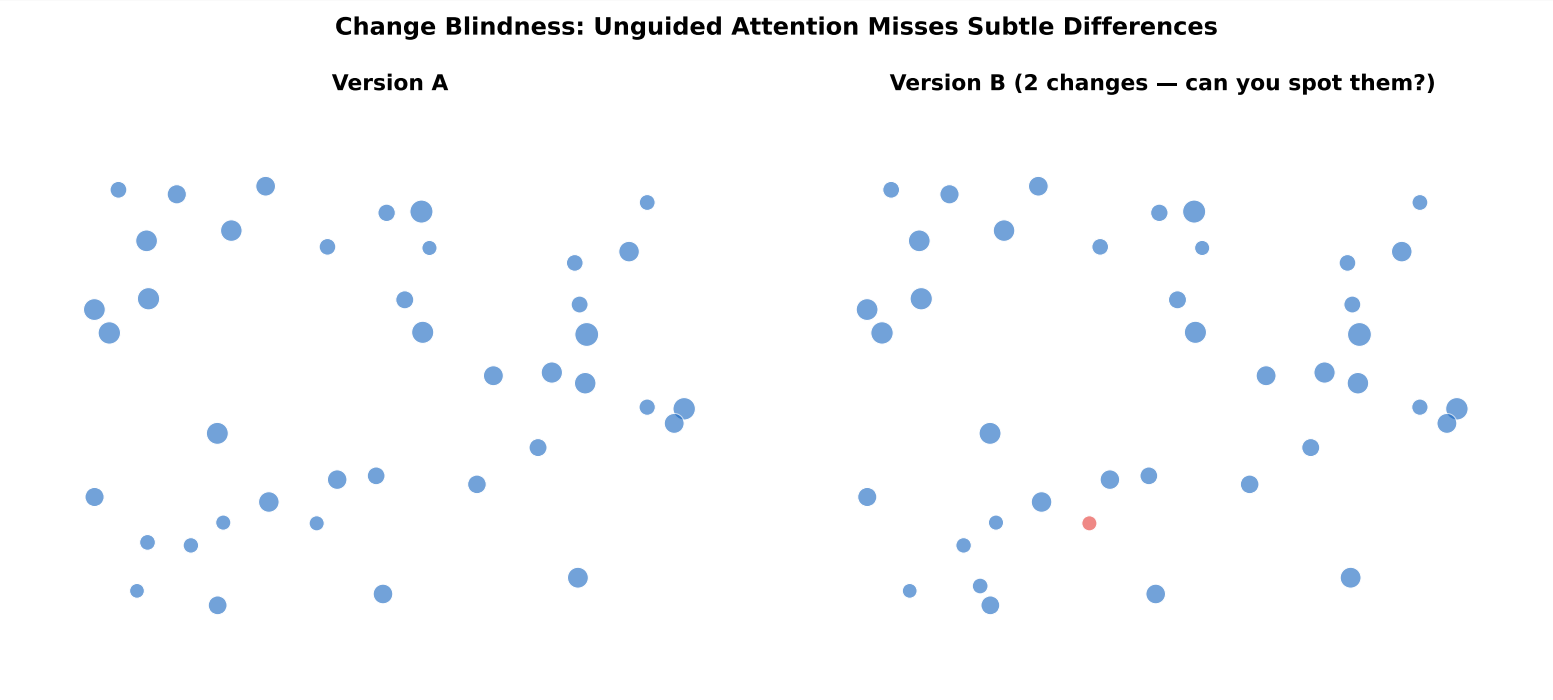
Cleveland & McGill (1984) showed experimentally that position judgments are $\sim 2\times$ more accurate than length judgments and $\sim 5\times$ more accurate than area judgments.

Weber's Law: Just-Noticeable Difference



Weber's law: $\Delta S/S = k$ (constant). The smallest detectable change is a fixed *proportion* of the stimulus, not a fixed amount. Implication: small differences in area or colour saturation are invisible; position differences are always detectable.

Change Blindness



Pro-Tip

When comparing two views (before/after, Version A/B), viewers miss subtle changes without direct annotation. Always highlight the specific data points that changed using colour, annotation, or connecting lines.

Choosing Channels: A Decision Framework

Data Type	Best Channel	Avoid
Quantitative	Position, length	Area, colour saturation
Ordinal	Position, size, saturation	Shape
Nominal ($\leq$7)	Colour hue, shape	Position (wastes axis)
Nominal ($>$7)	Faceting (small mult.)	Colour (too many hues)

Data Insight
Map the most important variable to the most effective channel. Secondary variables get secondary channels. Never encode the same variable redundantly on two channels unless deliberately reinforcing.

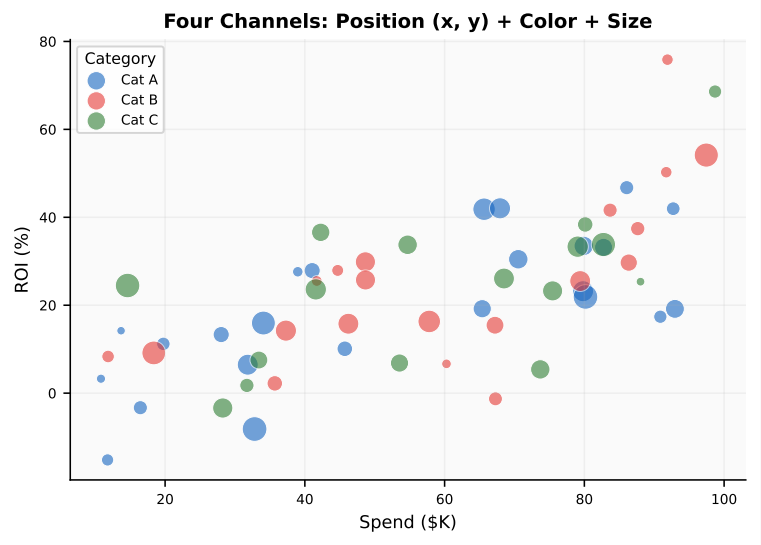
Encoding Channels in Code

Multi-Channel Encoding in R

```r
library(ggplot2)

# Four simultaneous channels:
# x = position (spend)
# y = position (ROI)
# color = category (nominal)
# size = budget (quantitative)

ggplot(df, aes(
    x = spend,
    y = roi,
    color = category,
    size = budget)) +
  geom_point(alpha = 0.6) +
  scale_color_manual(
    values = c(A = "#1565C0",
               B = "#E53935",
               C = "#2E7D32")) +
  scale_size_continuous(
    range = c(2, 10)) +
  theme_minimal() +
  labs(title = "4-Channel Plot")
```



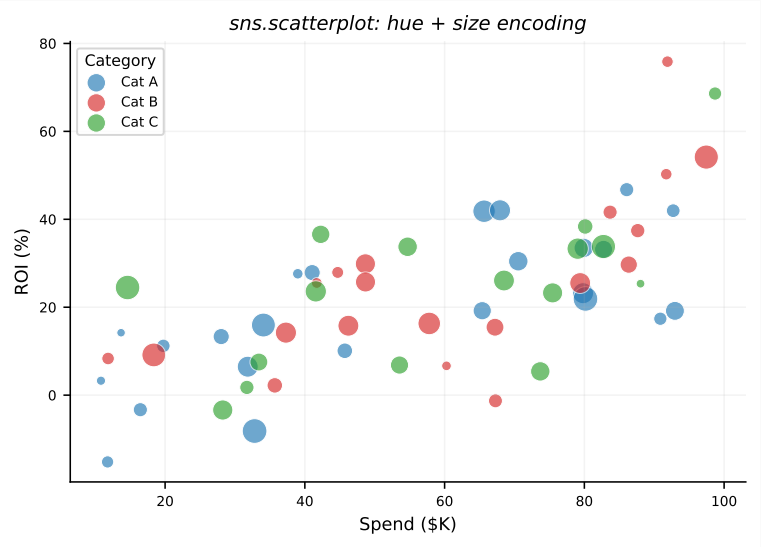
Multi-Channel Encoding in Python

```python
import seaborn as sns
import matplotlib.pyplot as plt

# seaborn handles channels via
# hue, size, style parameters

sns.scatterplot(
    data=df,
    x="spend",
    y="roi",
    hue="category",    # color
    size="budget",      # size
    alpha=0.6,
    palette={
        "A": "#1f77b4",
        "B": "#d62728",
        "C": "#2ca02c"},
    sizes=(30, 250),
)

plt.title("4-Channel Plot")
plt.tight_layout()
```



Gestalt-Aware Layout in R

```r
# Proximity: use faceting to
# separate groups spatially
ggplot(df, aes(x = x, y = y)) +
  geom_point(aes(color = group),
             size = 2, alpha = 0.6)
                  +
  facet_wrap(~group, ncol = 3) +
  theme_minimal() +
  theme(legend.position = "none")

# Enclosure: add shaded band
ggplot(df, aes(x = date, y = val))
    +
  annotate("rect",
    xmin = as.Date("2023-03-01"),
    xmax = as.Date("2023-06-01"),
    ymin = -Inf, ymax = Inf,
    fill = "#E3F2FD", alpha = 0.5)
        +
  geom_line(color = "#1565C0") +
  theme_minimal()
```

Design patterns:

- **Proximity** → faceting separates groups physically
- **Enclosure** → `annotate("rect")` highlights a time period
- **Connection** → `geom_line` links sequential points
- **Similarity** → consistent colour palette

Gestalt-Aware Layout in Python

```python
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3,
    sharey=True, figsize=(10, 3))

# Proximity via subplots
for ax, grp in zip(axes, groups):
    sub = df[df["group"] == grp]
    ax.scatter(sub["x"], sub["y"],
        c="#1565C0", s=25, alpha
            =0.6)
    ax.set_title(grp)

# Enclosure via axvspan
fig, ax = plt.subplots()
ax.axvspan("2023-03", "2023-06",
    color="#E3F2FD", alpha=0.5)
ax.plot(dates, values,
    color="#1565C0")

# Connection via plot()
ax.plot(x, y, "o-",
    color="#E53935", lw=1.5)
```

Python equivalents:

- **Proximity** → `subplots` with shared axes
- **Enclosure** → `ax.axvspan()` for shaded bands
- **Connection** → `plot("o-")` links markers
- **Similarity** → palette dict for consistent colours

Synthesis

Key Takeaways

1. **Pre-attentive attributes** (colour, shape, size, orientation) are detected in < 200 ms — use them for the most critical data distinction.
2. **Conjunction search** is slow ($\sim$50 ms/item) — never require viewers to combine two features to find a target.
3. **Gestalt laws** (proximity, similarity, continuity, closure, enclosure, connection) govern how elements are grouped — harness them or suffer misreadings.
4. **Channel effectiveness** (Cleveland & McGill, 1984): position $>$ length $>$ angle $>$ area $>$ colour — map the most important variable to the best channel.
5. **Weber's law**: the just-noticeable difference is proportional to the stimulus. Small data differences require high-precision channels (position), not low-precision ones (area, saturation).

Essential Readings for This Module

- Cleveland, W. S. & McGill, R. (1984). "Graphical Perception: Theory, Experimentation, and Application." *JASA*, 79(387).
- Healey, C. G. & Enns, J. T. (2012). "Attention and Visual Memory in Visualization." *IEEE TVCG*, 18(7).
- Ware, C. (2021). *Information Visualization: Perception for Design*, 4th ed., Chapters 1–5.
- Few, S. (2012). *Show Me the Numbers*, Chapters 4–5.
- Munzner, T. (2014). *Visualization Analysis and Design*, Chapter 5 (Marks and Channels).

Next Module: Color Theory & Accessibility (W01-M04)